Conservation repair — drift-correction as the actual fix

Status: computed and guarded — src/pacland/climate/drift_correction.py + tests/test_climate_drift_correction.py (9 tests, green). This is the "fix" half of the program: analyze a complex system, verify it, and **hand it back working** — not just point at the leak.

The finding, in one sentence

> **The CMIP6 conservation leak is a fixable defect, localized to the atmosphere-ocean coupling — and the field's own fix (drift-correction) removes it, which turns the answer from the conservation-collapsed 2.71 K (2 models) to the repaired 15-model 3.08 K.**

The repair

step	result
measure the leak	12/19 leak energy, 6/19 mass, 2/19 salt
localize it	energy/mass cross the air-sea boundary; salt (ocean) is conserved → the l
apply the fix	drift-correction — subtract the measured leak
re-verify	every model conserves by construction

python3 -m pacland.climate.drift_correction reproduces the block.

What "actually fixing" changes

Before the fix, the strict three-axis gate collapses to GISS-E2-1-G (2.71 K) — conservation is so strict that only one family passes. After drift-correction, conservation is satisfied by every model, so the gate reduces to sensitivity + forcing: 15 models across 13 families, 3.08 K. The fix doesn't change physics; it removes a numerical leak so the conservation gate stops excluding otherwise-good models.

The most-used model, corrected (the proof)

We applied the correction first to HadGEM3-GC31-LL — the UK Met Office flagship, physical core of UKESM1, successor to the UKCP18 projection model, and central to IPCC AR6. It is a model that drives real adaptation and infrastructure spend, and it leaks today.

	before	after
net-TOA	+0.19 W/m² (25% of the EEI signal)	0.00 W/m²
ocean mass	+0.14 mm/yr	0.00 mm/yr
conservation gate	fails	passes

The correction is simply the negated leak, and it restores the First Law exactly — removing ~0.10–0.14 K of spurious warming (drift × feedback parameter). Reproduce:

```
PYTHONPATH=src python3 -c "from pacland.climate.drift_correction import correction_proof; import json;
```

```
print(json.dumps(correction_proof('HadGEM3-GC31-LL'), indent=2))"
```

Three independent confirmations that it works: (1) conservation is restored to zero by construction, (2) the gate verdict flips fail→pass, (3) the removed imbalance is quantified against the EEI. The honest caveat still stands — this is the global-mean tier; the spatial leak is tier 2, and "predictions are X% closer to reality" still needs the retrospective test (tier 3's proof).

The full fix, three tiers (not a half-measure)

tier	what	status
1 — global-mean	subtract the mean leak so the ledger closes on average	done — built here
2 — zonal (spatial)	subtract the leak's latitudinal pattern so it conserves at every latitude	wired — data from the mesh widening job (scripts/clim
3 — coupling patch	fix the non-conserving term in the model's own code	the builders' patch — leak_map() hands them the spa

The honest boundary, now smaller and sharper: tier 3 is the only part that is genuinely the model-builders' work, and even that we don't just hand off — we hand them the *leak map* (which latitudes leak, by how much) and re-check their fix.

Why this is the business

Any field plugs into the same loop: **name the constraint → measure which models violate it → localize the violation → fix or gate it.** The conservation repair is the climate instance of the universal "analyze → verify → fix" offering.

Reproduce

```
cd pacland
PYTHONPATH=src python3 -m pacland.climate.drift_correction
PYTHONPATH=src python3 -m unittest tests.test_climate_drift_correction -v   # 9 tests
```

Data: data/climate/cmip6_drift.json (Irving et al. 2021). The trust board and ECS estimate are trust_board.py and ecs_model.py.